Lecture 23
Significance Tests

Review

1 What p* do we use to plan n when there is no earlier guess?

p* = 0.5: it makes p*(1 − p*) largest, so n is big enough whatever p is.

2 What is the null hypothesis H_0?

“No effect”: nothing interesting is happening, e.g. they are just guessing.

3 What is the alternative hypothesis H_A?

“An effect”: what we suspect is true, e.g. they can taste the difference.

Can you taste the difference?

Your roommate claims they can tell Coke from Pepsi. In a blind test, you pour one of the two at random, 8 times, and they name each cup.

8 of 8 cups named correctly

(1) Can they really tell, or were they just lucky? How could we decide?
(2) If they were only guessing, how likely is 8 for 8?
(3) What are the two ways to explain such a small probability?

(2) $(1/2)^8 = 1/256 \approx 0.004$
(3) Either something rare happened by chance, or "just guessing" is wrong
How rare is rare enough is our choice; by convention, below 0.05

Why test a hypothesis?

hypothesis

a claim about a population parameter (p or μ), e.g. p = 0.5: "they are just guessing"

purpose of a significance test

to decide whether the data give convincing evidence against a hypothesis

the logic

assume the null hypothesis is true, then ask how unlikely our data would be under that hypothesis

Significance tests

statistically significant

decidedly not due to ordinary variation: not chance, not a coincidence

significance test

uses data to decide whether a result is statistically significant

null hypothesis H_0

"no effect": nothing interesting is happening, e.g. they are just guessing

alternative hypothesis H_A

"an effect": what we suspect, e.g. they can taste the difference

Writing the hypotheses

θ and θ_0 ("theta" and "theta-naught")

θ is the parameter (p or μ); θ_0 is the value H_0 claims, e.g. $p_0 = 0.5$

$H_0: \theta = \theta_0$

H_0 always has the = sign, and is about a parameter, never $\hat{p}$ or $\bar{x}$

$H_A: \theta > \theta_0$, $\theta < \theta_0$ or $\theta \neq \theta_0$

upper-tailed, lower-tailed (one-sided) or two-sided; set before seeing data

Example: stating hypotheses

For each question, write H_A first (the suspicion), then H_0 (no effect). Give θ_0 and say which tail.

p = a population proportion; μ = a population mean

(a) Your roommate names Coke or Pepsi. Do they do better than guessing?
(b) A coin is spun on a table. Does it favor heads or tails?
(c) A chip bag says 28 g. Do the bags hold less than that on average?

(a) H_A: $p > 0.5$ H_0: $p = 0.5$ $p_0 = 0.5$, upper-tailed (p = chance of a right cup)
(b) H_A: $p \neq 0.5$ H_0: $p = 0.5$ $p_0 = 0.5$, two-sided (p = chance of heads)
(c) H_A: $\mu < 28$ g H_0: $\mu = 28$ g $\mu_0 = 28$ g, lower-tailed

Conditional probability

conditional probability P(A | B)

the probability that A happens, given that B is true; the bar | is read "given"

in a significance test

every probability is worked out given H_0 is true: P(8 of 8 | H_0 true) = $0.5^8 \approx 0.004$

Test statistics and P-values

test statistic z_{obs}

z_{obs} = (estimate − θ_0) / SE: how many SEs the estimate is from θ_0

P-value

the probability of observing a value as or more extreme than the test statistic, when H_0 is true
upper-tailed: $P(Z \geq z_{obs} \mid H_0 \text{ true})$

significance level α

the cutoff, chosen before collecting data; usually α = 0.05

the decision

P-value < α: reject H_0 (significant); P-value ≥ α: do not reject H_0